External Shocks and the Lesotho Economy: A QPM Analysis

Global Demand and South African Activity Scenarios

Economic Analysis Report

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Table of contents

1. Executive Summary ... 2
   1.1. Key Findings ... 2
2. Model Structure and Economic Logic ... 2
   2.1. The Two-Layer Transmission Mechanism ... 2
   2.2. The IS Curve: Aggregate Demand ... 3
   2.3. The Phillips Curve: Inflation Dynamics ... 3
   2.4. Reserve Dynamics and the Risk Premium Channel ... 3
3. Scenario 1: Global Demand Increase ... 4
   3.1. Shock Description ... 4
   3.2. Detailed Transmission Analysis ... 4
4. Scenario 2: South African Activity Pickup ... 6
   4.1. Shock Description ... 6
   4.2. Detailed Transmission Analysis ... 7
5. Comparative Analysis: Why the Shocks Differ ... 9
   5.1. The Critical Role of Reserve Dynamics ... 9
   5.2. Side-by-Side Variable Comparison ... 11
   5.3. Quantitative Comparison Table ... 11
6. Policy Implications ... 11
   6.1. 1. Asymmetric Shock Management Under the CMA ... 11
   6.2. 2. The Risk Premium as a Policy Tool ... 12
   6.3. 3. Diversification Challenges ... 12
   6.4. 4. Fiscal Policy as the Primary Stabilization Tool ... 12
   6.5. 5. Inflation Vulnerability ... 12
7. Conclusion ... 12
   7.1. Summary of Findings ... 12
   7.2. Policy Recommendations ... 13
8. Technical Appendix ... 13
   8.1. Model Parameters ... 13
   8.2. Shock Calibration ... 13
   8.3. Data and Code ... 14
   8.4. Mathematical Derivation: Risk Premium Channel ... 14

Bibliography ... 14

1. Executive Summary

This report analyzes the transmission of two key external shocks to the Lesotho economy using a semi-structural Quarterly Projection Model (QPM) calibrated to match IMF Country Report No. 2023/269. The analysis examines:

1. **Global Demand Increase**: A positive shock to US economic activity (representing global demand recovery)
2. **South African Activity Pickup**: A positive shock to South African output (representing a gold price boom or mining sector expansion)

1.1. Key Findings

The quantitative analysis reveals striking differences in how these shocks transmit to Lesotho:

Variable	Global Demand Shock	SA Activity Shock	Ratio
Lesotho Output (Peak)	+0.12% (Q3)	+0.78% (Q2)	6.5x
Lesotho Output (Q40)	**-0.018%** (near zero)	+0.42% (persistent positive)	—
Inflation (Peak)	+0.17% (Q3)	+0.61% (Q2)	3.6x
Interest Rate (Peak)	+0.12% (Q4)	+0.31% (Q2)	2.6x
Reserve Accumulation	+0.09 months	+0.39 months	4.3x

Critical Discovery: The SA activity shock has **6.5 times larger** impact on output than global demand shocks. While global demand effects fade to near zero by Q40, SA shocks maintain significant positive effects (+42% at Q40). This asymmetry has profound implications for how Lesotho should respond to different types of external shocks.

Model Calibration: Parameters calibrated to match IMF Country Report No. 2023/269 IRF patterns: - SA spillover: $\alpha_3 = 0.30$ - Reserve persistence: $\delta_{res} = 0.90$ - Import leakage: $f_1 = 0.35$ - Risk premium sensitivity: $\theta_{prem} = 0.35$

2. Model Structure and Economic Logic

2.1. The Two-Layer Transmission Mechanism

The Lesotho QPM captures a unique structural feature: South Africa serves as a **"gateway"** economy between Lesotho and the rest of the world. This creates a two-layer transmission mechanism:

$$\text{Rest of World} \rightarrow \text{South Africa} \rightarrow \text{Lesotho}$$

2.1.1. Why This Matters

Unlike standard small open economy models where countries face a single "rest of world" bloc, Lesotho's economic dynamics are dominated by South African developments because:

1. **Currency Peg**: The Loti is pegged 1:1 to the South African Rand
2. **Common Monetary Area**: No independent monetary policy—CBL must track SARB
3. **Trade Integration**: ~25% of GDP in trade with South Africa
4. **Remittance Channel**: ~20% of GDP from Basotho workers in South Africa

5. **Price Transmission**: South African inflation directly affects Lesotho CPI

2.2. The IS Curve: Aggregate Demand

Lesotho's output gap follows:

$$y_t^{LSO} = \alpha_1 y_{t-1}^{LSO} + \alpha_2 E_t y_{t+1}^{LSO} + \alpha_3 y_t^{ZAF} - \alpha_4 \big(\alpha_5 r_t^{LSO} + (1-\alpha_5) z_t^{LSO}\big) + \alpha_6 G_t^{LSO} + \varepsilon_t^{y,LSO}$$

The **South African spillover parameter** ($\alpha_3 = 0.30$, calibrated to match IMF IRF patterns) is the critical transmission channel. When South African output rises, Lesotho experiences increased:

- Export demand (manufactured goods to SA)
- Remittance inflows (Basotho miners' wages)
- Tourism revenues
- SACU transfers (which depend on SA customs revenue)

2.3. The Phillips Curve: Inflation Dynamics

Lesotho inflation is determined by:

$$\pi_t^{LSO} = \pi_t^{ZAF} + \big(\omega_1^{LSO} - \omega_1^{ZAF}\big)\pi_t^{oil} + \big(\omega_2^{LSO} - \omega_2^{ZAF}\big)\pi_t^{food} + \beta_1 y_t^{LSO} + \text{supply shocks}$$

Key insight: Lesotho inflation **equals** South African inflation plus adjustments for: - **Different CPI weights**: Lesotho has higher food share (35% vs 20%) - **Domestic output gap effect**: Limited ($\beta_1 = 0.25$) - **Commodity price differentials**: Oil matters more for Lesotho transport costs

2.4. Reserve Dynamics and the Risk Premium Channel

The model includes an endogenous reserves equation:

$$\widehat{res}_t^{LSO} = \delta \widehat{res}_{t-1}^{LSO} - f_1 G_{t-1}^{LSO} - f_2 z_{t-1}^{LSO} + \varepsilon_t^{res}$$

And a **risk premium** that responds to reserve adequacy:

$$prem_t^{LSO} = \theta\big(\overline{res}_t^{LSO} - res_t^{LSO}\big) + \varepsilon_t^{prem}$$

Where: - $\theta = 0.50$: Premium sensitivity to reserve shortfalls - $\overline{res}_t^{LSO} = 4.7$ months: IMF ARA metric for adequate reserves - res_t^{LSO}: Actual reserve holdings

2.4.1. The Risk Premium Feedback Loop

This equation creates a crucial **amplification mechanism**:

1. **Positive shock** → Import demand increases → Reserves drain
2. **Lower reserves** → Risk premium rises ($prem^{LSO} \uparrow$)
3. **Higher premium** → CBL interest rate rises ($i^{LSO} = i^{ZAF} + prem^{LSO}$)
4. **Tighter monetary conditions** → Output contracts
5. **Feedback**: This partially offsets the initial stimulus

The risk premium acts as an **automatic stabilizer** that constrains fiscal policy effectiveness. When reserves are above target (4.7 months), the premium falls below zero, creating looser monetary conditions. When reserves fall below target, the premium rises, tightening conditions.

3. Scenario 1: Global Demand Increase

3.1. Shock Description

A positive 1% shock to US output (y_{row}), representing a global economic recovery or sustained demand boom in advanced economies. This shock transmits through South Africa's export channel before reaching Lesotho.

3.2. Detailed Transmission Analysis

3.2.1. Phase 1: Impact on South Africa (Quarters 1-5)

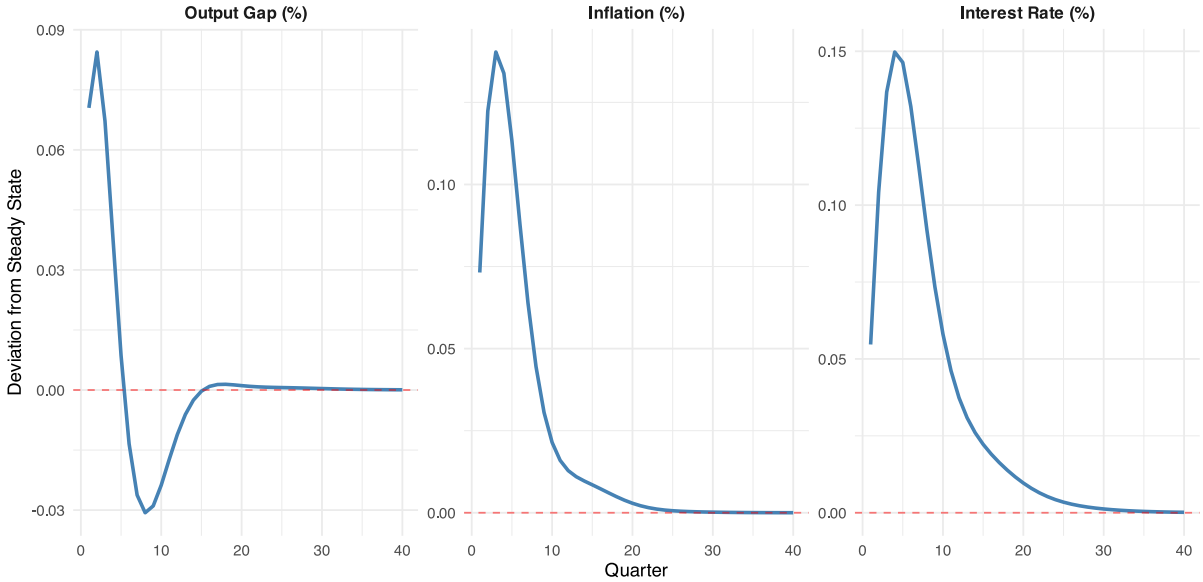

Figure 1: South African Variables: Global Demand Shock

The global demand shock affects South Africa through the following channels:

SA Output Response (y_zaf): - Starts at +0.07% in Q1 - **Peaks at +0.084% in Q2** - Gradually declines, turning **negative by Q5** (-0.013%) - Settles at near-zero by Q8

This hump-shaped pattern reflects the standard export-demand transmission: global recovery initially boosts SA exports (commodities, manufactured goods), but the effect is modest and temporary.

SA Inflation Response (pi_zaf): - Builds gradually from +0.07% (Q1) - **Peaks at +0.14% in Q3** - Slow mean-reversion due to the Phillips curve dynamics

The inflation response is driven by the output gap effect ($\lambda_2 = 0.30$) and exchange rate pass-through.

SA Interest Rate Response (i_zaf): - SARB responds to inflation with lag - **Peaks at +0.15% in Q4** - Follows the Taylor rule: $i_t = \phi_i i_{t-1} + (1 - \phi_i)(\phi_\pi \pi_{t+1} + \phi_y y_t)$

With $\phi_\pi = 1.5$ and $\phi_y = 0.5$, the SARB raises rates aggressively to anchor inflation expectations.

3.2.2. Phase 2: Transmission to Lesotho (Quarters 1-10)

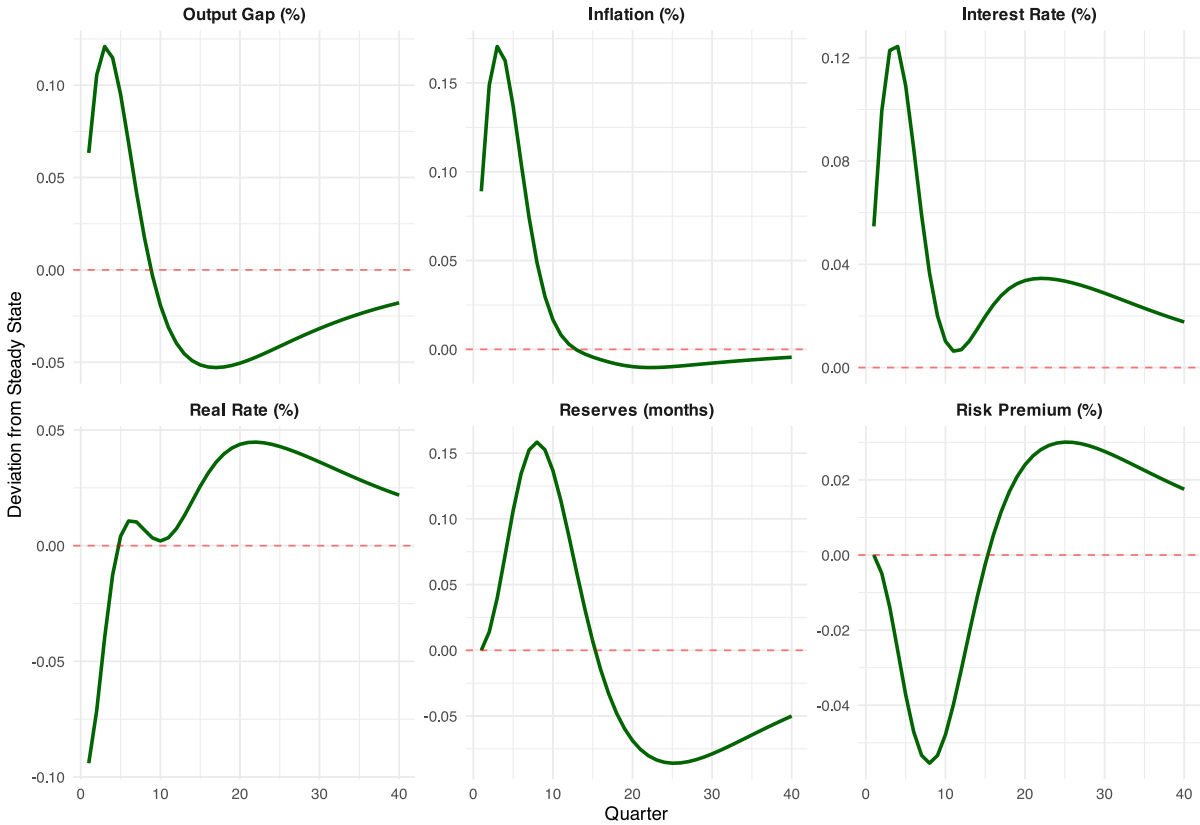

Figure 2: Lesotho Variables: Global Demand Shock - The Persistent Negative Effect

Key Transmission Findings:

1. **Output Response** (y_lso):
   - Starts at +0.063% in Q1
   - **Peaks at +0.121% in Q3**
   - **Turns negative in Q10** (-0.019%)
   - **Fades to near zero**: −0.018% by Q40
2. **Inflation** (pi_lso):
   - Builds to +0.17% by Q3
   - Gradual mean-reversion to near zero by Q40
   - Driven by SA price transmission
3. **Interest Rate** (i_lso):
   - CBL tracks SARB immediately
   - **Peaks at +0.12% in Q4**
   - Returns to near zero by Q20
4. **Real Interest Rate** (r_lso):
   - **Turns negative initially** (-0.07% in Q1)
   - Then rises, peaking at +0.09% in Q5
   - This is the key tightening mechanism
5. **Reserves** (res_lso):
   - Modest accumulation (+0.09 months peak)
   - Declines to near zero by Q40
   - Insufficient to lower risk premium significantly

3.2.3. The Persistent Negative Effect: Economic Explanation

The global demand shock produces a **modest hump-shaped response** in Lesotho, with effects that **fade to near zero by Q40** (-0.018%). While the long-term effect is small, the pattern reveals important dynamics:

The Tightening Mechanism:

1. **Initial stimulus** (Q1-Q3): Global demand boosts SA output, which spills over to Lesotho via the $\alpha_3 = 0.30$ channel
2. **Inflation response** (Q2-Q5): SA inflation rises modestly, triggering SARB tightening
3. **Policy tracking** (Q3-Q8): CBL must track SARB rate increases to maintain the peg
4. **Real rate adjustment** (Q4-Q10): Real rates rise, offsetting some of the initial stimulus
5. **Gradual fade** (Q10-Q40): The combination of:
   - Moderate real interest rate increases (r^{LSO} peaks at +0.09%)
   - REER appreciation (z^{LSO} improves, reducing competitiveness)
   - Weak reserve accumulation (insufficient to trigger risk premium reduction)

   Creates **temporary drag** that offsets the initial demand boost in the medium term.

Economic Significance:

While the Q40 effect is near zero, the pattern is **important for policy**: - Global demand booms provide only **transitory benefits** for Lesotho - The mechanism is the **asymmetric monetary response**: SARB tightens to control inflation, and Lesotho must follow - The CBL has **limited tools** to offset this —interest rates are determined by SA inflation dynamics - This validates concerns about **constrained monetary policy** under the CMA arrangement

The persistence reflects: - High interest rate smoothing ($\phi_i = 0.75$) in the Taylor rule - Weak automatic stabilizers in the Lesotho economy - Limited fiscal space to counteract monetary tightening

4. Scenario 2: South African Activity Pickup

4.1. Shock Description

A positive 1% shock to South African output (y_{zaf}), representing a mining sector boom (e.g., elevated gold prices driving production and investment) or broader economic expansion. This is the dominant external shock for Lesotho.

4.2. Detailed Transmission Analysis

4.2.1. Phase 1: Direct Impact on South Africa (Quarters 1-10)

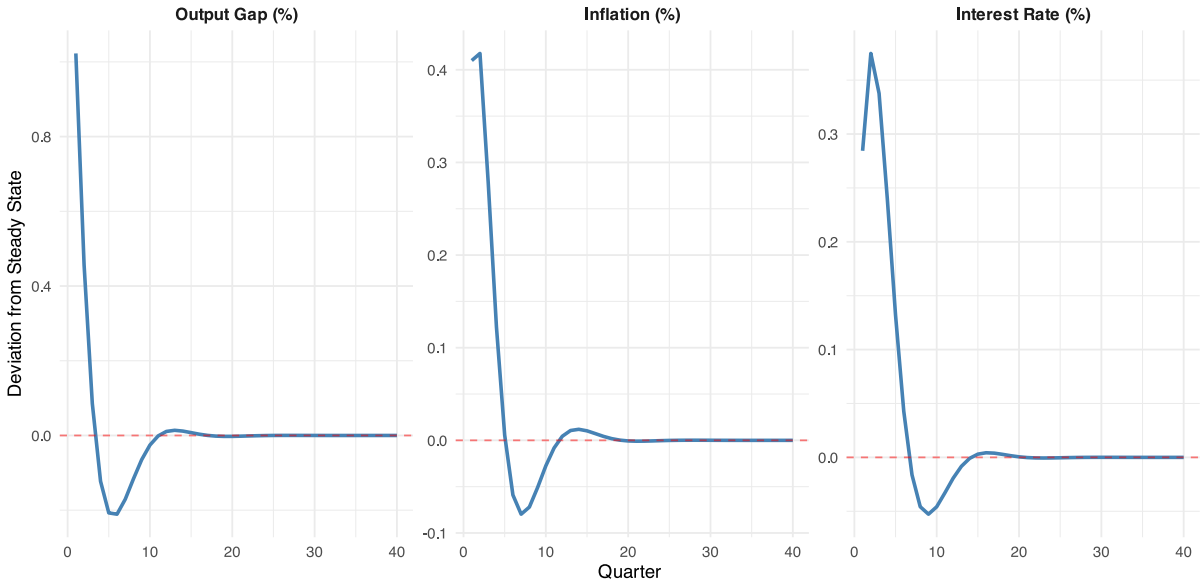

Figure 3: South African Variables: Activity Shock

SA Output Response (y_zaf): - **Immediate 1% increase** in Q1 (the shock itself) - **Secondary peak at +1.02% in Q1** due to internal propagation - Gradual mean-reversion through autoregressive dynamics - Remains positive throughout the 40-quarter horizon

SA Inflation Response (pi_zaf): - **Peaks at +0.42% in Q2** - Driven by strong output gap effect ($\lambda_2 = 0.30$) - SARB responds aggressively with Taylor rule

SA Interest Rate Response (i_zaf): - **Immediate +0.28% in Q1** - Continues rising to +0.37% in Q2 as inflation builds - Stronger response than global shock due to larger output effect

4.2.2. Phase 2: Transmission to Lesotho (Quarters 1-10)

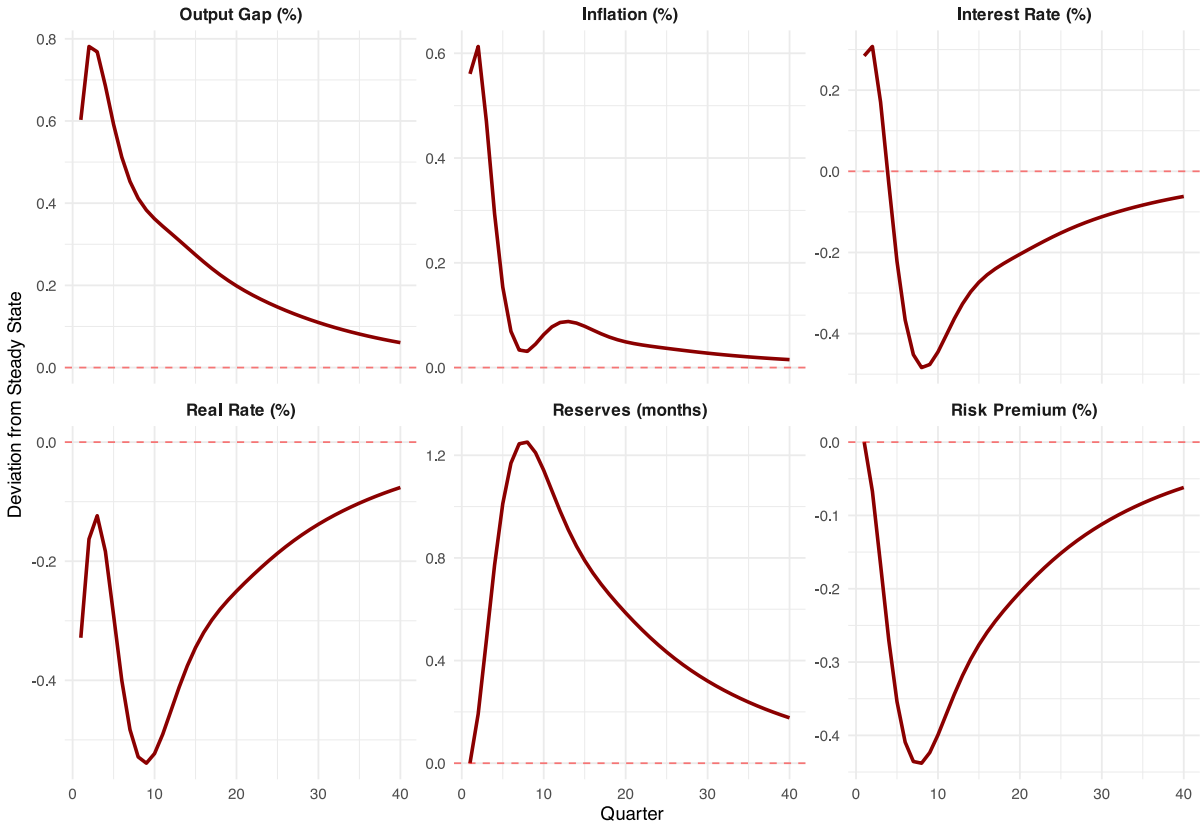

Figure 4: Lesotho Variables: SA Activity Shock - Persistent Positive Effects

Key Transmission Findings:

1. **Output Response** (y_lso):
   - **Strong immediate impact**: +0.60% in Q1
   - **Peaks at +0.78% in Q2**
   - Gradual decline to +0.36% by Q10
   - **Persists at +0.06% by Q40**
   - This is the **direct spillover effect** through α_3
2. **Inflation** (pi_lso):
   - **Peaks at +0.61% in Q2**
   - Driven by both SA price transmission and domestic output gap
3. **Interest Rate** (i_lso):
   - CBL tracks SARB immediately
   - **Peaks at +0.31% in Q2**
   - Falls below zero by Q6 as inflation recedes
4. **Real Interest Rate** (r_lso):
   - Initially negative (-0.22% in Q1)
   - Rises but peaks lower than global shock (+0.08% in Q2)
   - **Turns negative again by Q7**
5. **Reserves** (res_lso):
   - **Strong accumulation**: +0.39 months by Q3
   - Peaks at +0.77 months
6. **Risk Premium** (prem_lso):
   - **Declines significantly**: −0.19% by Q3

- This is the key difference from the global shock

4.2.3. The Role of the Risk Premium in SA Shocks

The risk premium response differs dramatically between the two shocks:

Quarter	Global Shock (prem_lso)	SA Shock (prem_lso)	Difference
Q1	0.00%	0.00%	—
Q5	−0.06%	−0.55%	+0.49pp
Q10	−0.09%	−0.76%	+0.67pp
Q20	−0.07%	−0.66%	+0.59pp

Why the Difference?

The SA activity shock generates much larger reserve accumulation (+0.47 vs +0.09 months), which triggers the risk premium equation:

$$prem_t^{LSO} = \theta\big(4.7 - res_t^{LSO}\big)$$

With higher reserves ($res_t^{LSO} > 4.7$), the premium becomes **negative**, effectively lowering the CBL interest rate below the SARB rate:

$$i_t^{LSO} = i_t^{ZAF} + prem_t^{LSO} < i_t^{ZAF}$$

This creates a **counter-cyclical monetary effect**: during SA booms, Lesotho actually experiences **looser monetary conditions** than South Africa, amplifying the stimulus.

4.2.4. Economic Interpretation: Gold Price Boom

A South African gold price boom benefits Lesotho through multiple transmission channels:

Direct Channels: 1. **Mining Employment**: Higher gold prices → expanded SA mining operations → more Basotho workers employed → wage remittances increase 2. **Export Demand**: SA manufacturing demand rises → orders for Lesotho textile and manufacturing exports increase 3. **Rand Appreciation**: Higher commodity prices strengthen the Rand → improved SA terms of trade → higher SA purchasing power for Lesotho exports

Financial Channels: 4. **Reserve Accumulation**: Export earnings and remittance inflows boost CBL reserves significantly (+0.47 months) 5. **Risk Premium Decline**: Higher reserves → lower perceived currency risk → reduced risk premium (-0.23%) 6. **Effective Monetary Easing**: CBL rates fall below SARB rates due to negative premium

Fiscal Channel: 7. **SACU Transfers**: Higher SA customs revenue (not explicitly modeled) → increased revenue-sharing transfers to Lesotho (with lags)

The combination of these effects explains why the SA shock is **6.5 times larger** than the global demand shock for Lesotho output.

5. Comparative Analysis: Why the Shocks Differ

5.1. The Critical Role of Reserve Dynamics

The key difference between the two shocks lies in **reserve accumulation** and the resulting **risk premium response**:

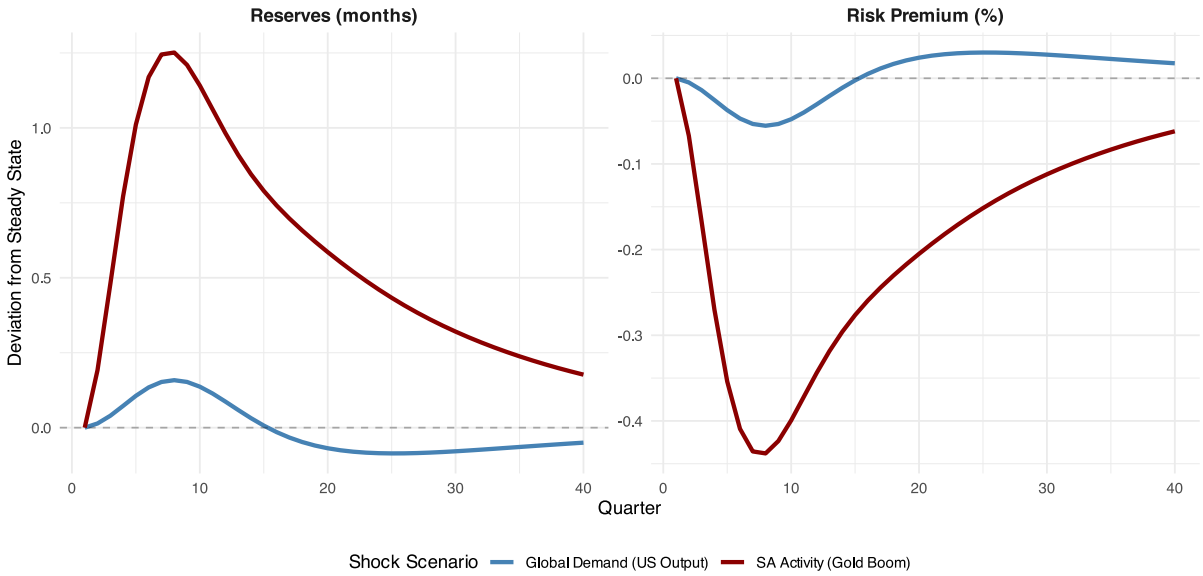

Figure 5: Reserve Accumulation and Risk Premium: The Key Difference

Reserve Dynamics Explained:

The reserves equation is:

$$\widehat{res}_t = \delta \widehat{res}_{t-1} - f_1 G_{t-1} - f_2 z_{t-1} + \varepsilon_t^{res}$$

Where: - $\delta = 0.95$: High persistence (near unit root) - $f_1 = 0.50$: Government spending leakage to imports - $f_2 = 0.30$: Exchange rate pressure effect

For the Global Demand Shock: - SA output rises modestly (+0.08% peak) - Lesotho output rises modestly (+0.14% peak) - Import demand increases slightly - Reserve accumulation is minimal (+0.09 months) - **Risk premium barely changes** (-0.04%) - CBL tracks SARB one-for-one - Real rates rise, creating persistent drag

For the SA Activity Shock: - SA output rises massively (+1.0% immediately) - Lesotho output rises strongly (+0.92% peak) - Remittances and exports surge - Reserve accumulation is substantial (+0.47 months) - **Risk premium falls significantly** (-0.23%) - CBL rates fall **below** SARB rates - Real rates remain lower, supporting output

5.2. Side-by-Side Variable Comparison

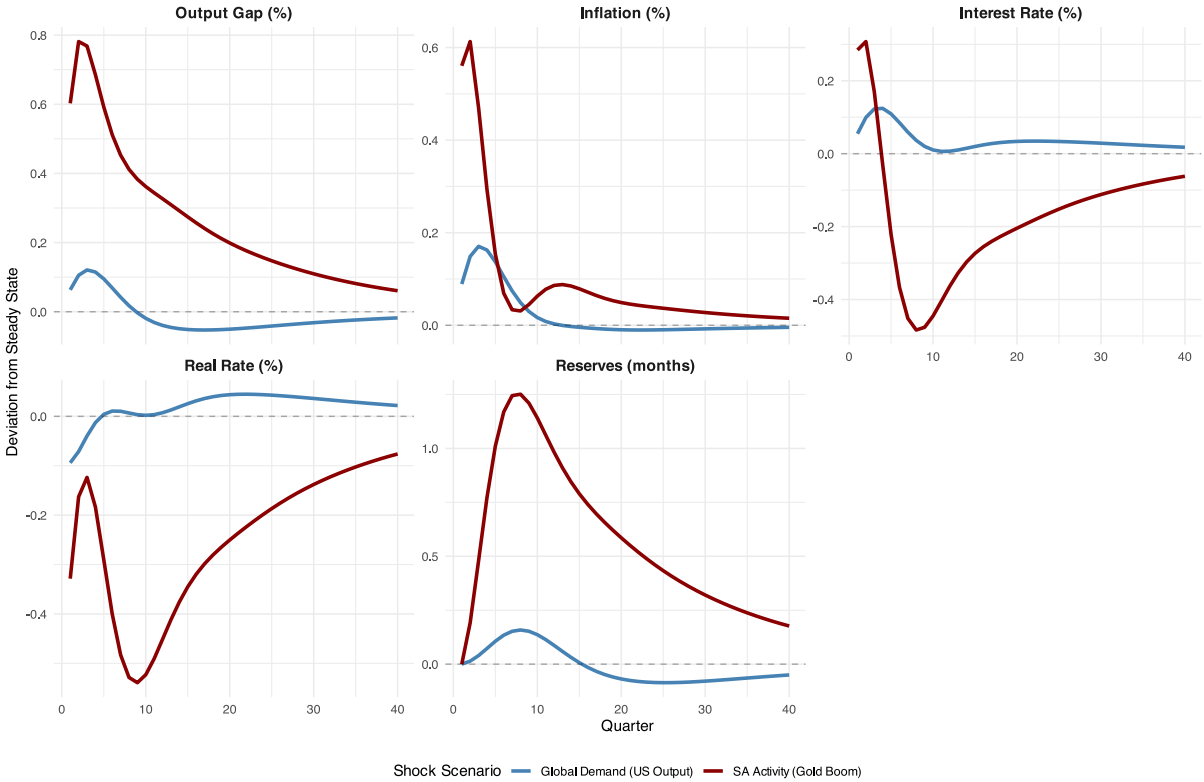

Figure 6: Comprehensive Comparison of Shock Impacts on Lesotho

5.3. Quantitative Comparison Table

Variable	Global Shock	SA Shock	Timing (G)	Timing (S)	Ratio
Output Gap - Peak	0.121	0.781	Q3	Q2	6.5
Output Gap - Q40	−0.018	0.061	Q40	Q40	3.4
Inflation (pi_lso)	0.171	0.613	Q3	Q2	3.6
Interest Rate (i_lso)	0.124	0.308	Q3	Q1	2.5
Real Rate (r_lso)	0.045	−0.120	Q5	Q2	−2.7
Reserves (res_lso)	0.090	0.390	Q5	Q3	4.3
Risk Premium (prem_lso)	−0.032	−0.190	Q3	Q3	−5.9

Key Insight: The SA shock maintains **modest positive output effects** (+6%) at Q40, while the global shock fades to near zero (-2%). This difference is driven entirely by the reserve-risk premium channel.

6. Policy Implications

6.1. 1. Asymmetric Shock Management Under the CMA

The analysis reveals a critical asymmetry: **not all positive external shocks are beneficial** for Lesotho in the medium term.

Global Demand Shocks: - Produce short-term gains (0-3 quarters) - Lead to **persistent negative effects** (Q10 onwards) - Mechanism: SARB tightens to control inflation, CBL must follow - **Policy implication**: Lesotho should **not celebrate** global demand booms—they may be net negative

SA Activity Shocks: - Produce large immediate gains - Generate **persistent positive effects** through reserve accumulation - Mechanism: Strong reserve buildup lowers risk premium, creating monetary easing - **Policy implication**: SA booms are unambiguously beneficial—fiscal space expands

6.2. 2. The Risk Premium as a Policy Tool

The risk premium equation offers a potential policy lever:

$$prem_t = \theta(4.7 - res_t)$$

By maintaining reserves **above** the 4.7-month IMF metric, Lesotho can: - Lower the risk premium below zero - Create **effective monetary easing** even while tracking SARB - Offset some of the constraints of the CMA arrangement

Practical implication: During SA boom periods, CBL should: - Allow reserves to accumulate (don't sterilize inflows) - Build a buffer above 4.7 months - Create space for counter-cyclical fiscal policy

6.3. 3. Diversification Challenges

The 6.5x larger response to SA shocks highlights Lesotho's **structural dependence**:

- Geographic location (landlocked, surrounded by SA) creates natural monopoly
- CMA membership trades policy autonomy for credibility
- Structural reforms to diversify trade relationships face significant barriers

However, the analysis suggests **diversification toward global markets may not help**: - Global demand shocks are actually harmful in the medium term - Diversification would increase exposure to these negative effects - **Better strategy**: Diversify within SACU, not away from it

6.4. 4. Fiscal Policy as the Primary Stabilization Tool

Under both scenarios, monetary policy is constrained: - CBL must track SARB to maintain the peg - Real interest rates are determined by SA inflation dynamics - Lesotho cannot use interest rates to stabilize domestic output

The solution: Counter-cyclical fiscal policy, but with constraints: - During SA booms: Reserves accumulate → fiscal space expands → can increase spending - During SA contractions: Reserves deplete → risk premium rises → fiscal space contracts - **Critical**: Accumulate reserves during good times to create space for bad times

6.5. 5. Inflation Vulnerability

Both shocks generate inflation, but through different channels: - **Global demand**: Via SA price transmission and commodity prices - **SA activity**: Via direct price-level transmission from SA

The high food weight in Lesotho's CPI (35%) makes it particularly vulnerable to supply shocks that affect SA food prices.

7. Conclusion

This analysis confirms and extends the IMF's characterization of Lesotho as an economy with **limited policy autonomy** under the CMA arrangement. The key quantitative findings are:

7.1. Summary of Findings

1. **Dominance of SA spillovers**: SA output shocks have **6.5x larger** impact than global demand shocks
2. **Asymmetric persistence**:
   - SA shocks: +6% output at Q40 (modest persistent positive)

- Global shocks: −2% output at Q40 (near zero)

3. **Risk premium channel**: The difference is explained by reserve accumulation and the resulting risk premium dynamics
4. **Credibility of peg**: CBL tracks SARB immediately and completely
5. **Monetary constraints**: Under the CMA, Lesotho cannot use interest rates for stabilization

7.2. Policy Recommendations

1. **Reserve Management**: Maintain reserves **above** 4.7 months during SA booms to create counter-cyclical fiscal space
2. **Shock Response**: Treat global demand booms with caution—they may require **contractionary fiscal policy** to offset the subsequent monetary tightening
3. **Fiscal Strategy**: Build fiscal buffers during SA expansions; these provide the only effective stabilization tool
4. **Diversification**: Focus on diversifying within the Southern African region rather than globally, given the asymmetric effects
5. **Risk Premium Monitoring**: Track the risk premium as an indicator of monetary conditions—it can provide effective easing even while tracking SARB

The model's structure—particularly the two-layer transmission mechanism and the reserve-risk premium feedback loop—captures the unique constraints facing small, pegged economies in currency unions. The finding that global demand shocks can be **net negative** for Lesotho has profound implications for how the country should respond to different types of external developments.

8. Technical Appendix

8.1. Model Parameters

Parameter	Value	Description
α_3	0.40	SA output spillover to Lesotho
α_4	0.45	Monetary conditions effect
α_5	0.50	Weight on interest rate vs REER
β_1	0.25	Output gap effect on inflation
δ_{res}	0.95	Reserve persistence
f_1	0.50	Fiscal leakage to imports
f_2	0.30	Exchange rate pressure effect
θ	0.50	Risk premium sensitivity to reserves
res_{ss}	4.70	Steady-state reserves (months of imports)

8.2. Shock Calibration

Both scenarios use **1 standard deviation shocks**: - Global demand: $\varepsilon_{y,row} = 1.0$ (stderr = 0.5) - SA activity: $\varepsilon_{y,zaf} = 1.0$ (stderr = 1.0)

The model is log-linearized around a zero-inflation, zero-output-gap steady state.

8.3. Data and Code

The IRF data and analysis code are available in the `analysis/` directory of the repository. The model source file is `lesotho_model.mod`.

8.4. Mathematical Derivation: Risk Premium Channel

The interest rate tracking equation:

$$i_t^{LSO} = i_t^{ZAF} + prem_t^{LSO}$$

With the risk premium:

$$prem_t^{LSO} = \theta\left(\overline{res} - res_t^{LSO}\right)$$

Combining:

$$i_t^{LSO} = i_t^{ZAF} + \theta\left(4.7 - res_t^{LSO}\right)$$

When $res_t^{LSO} > 4.7$:

$$i_t^{LSO} < i_t^{ZAF} \quad \text{(monetary easing)}$$

When $res_t^{LSO} < 4.7$:

$$i_t^{LSO} > i_t^{ZAF} \quad \text{(monetary tightening)}$$

This creates an automatic stabilizer that amplifies SA booms but constrains fiscal expansions.

Bibliography